Worked solutions to

Bayesian Statistical Methods

With Applications to Machine Learning — Reich & Ghosh (2nd ed.)

Chapter 6

Model selection and diagnostics

Every exercise is presented in the same order:

- the **exact** exercise text from the book;
- the problem **in plain words**;
- the **background** you need, recalled from scratch;
- the **key idea** that unlocks the problem;
- a **step-by-step solution** — every algebra step written out, with a *Why?* note whenever a step is not obvious;
- a **sanity check** of the answer;
- the boxed **answer**;
- **takeaways** and common mistakes.

All numbers and figures are reproduced by `ch06/code/ch06.R` (base R + MASS + splines).

Sai Sridhar · October 20, 2026

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Answers at a glance

1	5-fold CV clearly prefers M_2 (RMSE 22.0 vs 31.3; log score -500 vs -541).
2	M_2 fails PPCs for skewness, maximum and negative values $\Rightarrow$ model $\log(\text{ozone})$ or use a gamma likelihood.
3	DIC/WAIC slightly prefer separate rates (≈ 16 vs 17) and are essentially unchanged for $c = 1$ vs 10.
4	Collinear markers: individual effects uncertain, sum precise; posterior spread over overlapping subsets; summarise by region.
5	Logit and probit are indistinguishable (DIC 2525 vs 2527; WAIC 2525 vs 2527).
6	Logistic model under-predicts between-village variation (PPP ≈ 0) $\Rightarrow$ add village random effects.
7	LOO prefers M_1 (log score -23.2 vs -24.2); $\beta_2 \approx 1$ anyway.
8	M_2 passes the PPCs (PPP 0.63); no player is clearly unusual.
9	DIC and WAIC both choose $L = 4$.
10	$L = 2$ leaves autocorrelated residuals (PPP 0.016) $\Rightarrow$ larger L or a model for differences.
11	$\pi(c) = \frac{n}{(1+nc)^2}$, i.e. $c = \frac{U}{n(1-U)}$ with $U \sim \text{Uniform}(0, 1)$.
12	DIC/WAIC pick $J = 24$, CV picks $J = 18$ (flat for $J = 10$–24); residuals autocorrelated.
13	t model passes all four checks; Gaussian fails kurtosis (PPP 0.002).
14	BF: coherent but prior-sensitive; SSVS: many models but needs care with collinearity; CV: assumption-light but costly; DIC/WAIC: cheap but approximate.

A five-minute primer on the tools of this chapter

Background you need

1. Cross-validation (CV). Split the data into K folds; fit the model without fold k and predict fold k; repeat for all folds. Score the predictions by mean squared error or by the *log predictive density* $\sum_i \log p(Y_i \mid \mathbf{Y}_{-\text{fold}(i)})$, estimated by $\log \frac{1}{S}\sum_s f(Y_i \mid \theta^{(s)})$. Leave-one-out (LOO) is $K = n$.

2. DIC. With deviance $D(\theta) = -2\log f(\mathbf{Y} \mid \theta)$:

$$\text{DIC} = \bar{D} + p_D, \qquad \bar{D} = \text{E}[D(\theta) \mid \mathbf{Y}], \qquad p_D = \bar{D} - D(\hat{\theta}) \;\; (\hat{\theta} = \text{posterior mean}).$$

p_D is the *effective* number of parameters. Smaller DIC is better.

3. WAIC. With pointwise draws $\log f(Y_i \mid \theta^{(s)})$:

$$\text{WAIC} = -2\sum_i \log \overline{f}_i + 2p_W, \qquad \overline{f}_i = \frac{1}{S}\sum_s f(Y_i \mid \theta^{(s)}), \qquad p_W = \sum_i \text{Var}_s[\log f(Y_i \mid \theta^{(s)})].$$

It approximates LOO-CV. Smaller is better; a difference of $\gtrsim 5$ is usually meaningful.

4. Posterior predictive checks (PPC). For each posterior draw, simulate a replicate dataset $\mathbf{Y}^{\text{rep}}$ of the same size. For a statistic T (mean, max, skewness, autocorrelation, …), compare $T(\mathbf{Y})$ with the distribution of $T(\mathbf{Y}^{\text{rep}})$:

$$\text{PPP} = \text{Prob}\big(T(\mathbf{Y}^{\text{rep}}) \geq T(\mathbf{Y}) \mid \mathbf{Y}\big).$$

A PPP near 0 or 1 means the model cannot reproduce that feature of the data. Choose statistics that target plausible failures and are *not* directly fitted by the model.

Exercise 6.1 — Ozone: one predictor or three?

Exercise 1 (exact text from the book)

Download the airquality dataset in R. Compare the following two models using 5-fold cross validation:

$$M_1 : \text{ozone}_i \sim \text{Normal}(\beta_1 + \beta_2\, \text{solar.R}_i,\ \sigma^2)$$
$$M_2 : \text{ozone}_i \sim \text{Normal}(\beta_1 + \beta_2\, \text{solar.R}_i + \beta_3\, \text{temp}_i + \beta_4\, \text{wind}_i,\ \sigma^2).$$

Specify and justify the priors you select for both models.

In plain words

Predict daily ozone in New York either from solar radiation alone, or from solar radiation, temperature and wind; compare the models by how well they predict held-out days.

Background you need

Bayesian linear regression and its Gibbs sampler (Chapter 4 primer). K-fold CV and the log predictive density (primer, point 1).

Key idea

Use the *same* folds for both models, so the comparison is fair, and score both by squared error and by log predictive density (which also rewards honest uncertainty).

Solution

Data.

111 days with all four variables recorded (days with missing ozone or radiation dropped). Mean ozone 42.1 ppb, SD 33.3.

Priors and justification.

Covariates standardised; $\beta_1, \beta_j \sim \text{N}(0, 100^2)$ (an SD of 100 ppb per SD of a covariate is far wider than any plausible effect, so the priors are effectively uninformative), $\sigma^2 \sim \text{InvGamma}(0.1, 0.1)$ (vague). The same priors are used in both models, so differences come from the data.

5-fold CV

(random folds; for each fold fit by Gibbs on 4/5 of the days and predict the rest):

	RMSE (ppb)	MAE (ppb)	log predictive score
M_1	31.3	24.4	-540.7
M_2	**22.0**	**16.4**	$\mathbf{-500.3}$

Full-data fits

(coefficients per SD):

	M_1 mean (95% CI)	M_2 mean (95% CI)
solar.R	11.6 (5.8, 17.5)	5.4 (1.3, 9.7)
temp	—	15.8 (11.0, 20.5)
wind	—	-11.8 $(-16.4, -7.2)$
σ	31.5	21.3

Answer

M_2 predicts much better: CV RMSE 22.0 vs. 31.3 ppb and log predictive score -500 vs. -541 (a difference of 40 log units is overwhelming). Hot, calm days have much higher ozone; temperature and wind matter more than solar radiation.

Takeaway & common mistakes

The log score difference (40) is the CV analogue of a log Bayes factor — it is huge here. But "better" is not "good": Exercise 2 shows M_2 is still misspecified.

Exercise 6.2 — Posterior predictive checks for the ozone model

Exercise 2 (exact text from the book)

Fit model M_2 to the airquality data from the previous problem, and use posterior predictive checks to verify that the model fits well. If you find model misspecification, suggest (but do not fit) alternatives.

In plain words

Simulate fake ozone datasets from the fitted M_2 and see whether they look like the real data.

Background you need

PPC: primer, point 4. Ozone concentrations are positive and right-skewed.

Key idea

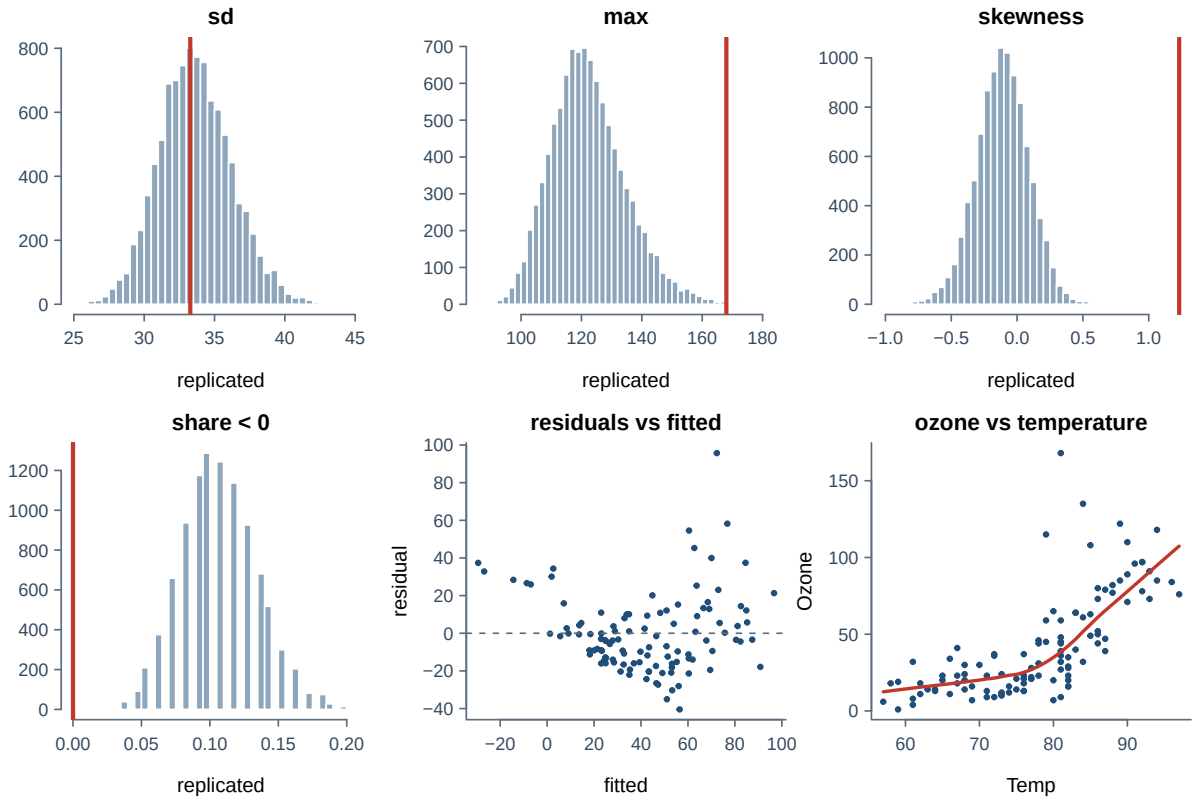
Normal linear models are symmetric and can go negative; test statistics that target skewness, extremes and positivity will reveal whether that matters here.

Solution

Replicates.

For each of 10,000 posterior draws, $Y_i^{\text{rep}} = \mathbf{X}_i\boldsymbol{\beta}^{(s)} + \sigma^{(s)}\varepsilon_i$.

Statistic T	observed	replicated mean	replicated 95% interval	PPP
mean	42.1	42.1	(36.5, 47.7)	0.50
SD	33.3	33.7	(28.9, 39.1)	0.54
minimum	1.0	-47.5	$(-80.6,\ -22.6)$	**0.000**
maximum	168	122.5	(101.5, 150.2)	**0.002**
skewness	1.23	-0.11	$(-0.51,\ 0.27)$	**0.000**
share < 0	0	0.106	(0.054, 0.162)	**1.000**



› Diagnosis.

Mean and SD are fine (the model fits them directly), but the replicates are **symmetric** (skewness ≈ 0 vs. 1.23), never reach the observed **maximum** (168), and about **11% of replicated days have negative ozone**. The residual and scatter plots show curvature (ozone rises sharply above ≈ 80°F) and a spread that grows with the fitted value.

› Suggested alternatives.

- Model $\log(\text{ozone})$ (or a gamma / log-normal likelihood with a log link): positive, right-skewed, variance growing with the mean.
- Add nonlinear terms: a quadratic or spline in temperature, and a temperature $\times$ wind interaction.
- Check residual autocorrelation over consecutive days (weather persists).

Answer

M_2 reproduces the mean and SD but fails for the minimum, maximum, skewness and positivity (PPP ≈ 0 or 1): it is misspecified. Use a log transform or a gamma likelihood with log link, and allow a nonlinear temperature effect and a temperature–wind interaction.

Takeaway & common mistakes

Choose PPC statistics the model does *not* fit by construction; the mean and SD will almost always "pass".

Exercise 6.3 — Mr. October with DIC and WAIC

Exercise 3 (exact text from the book)

Use the "Mr. October" data (Section 2.5) $Y_1 = 563$, $N_1 = 2820$, $Y_2 = 10$ and $N_2 = 27$. Compare the two models:

$$M_1 : Y_1|\lambda_1 \sim \text{Poisson}(N_1\lambda_1) \text{ and } Y_2|\lambda_2 \sim \text{Poisson}(N_2\lambda_2)$$
$$M_2 : Y_1|\lambda_0 \sim \text{Poisson}(N_1\lambda_0) \text{ and } Y_2|\lambda_0 \sim \text{Poisson}(N_2\lambda_0).$$

using DIC and WAIC. Assume the Uniform(0,c) prior for all λ_j, and compare the results for $c = 1$ and $c = 10$.

In plain words

Redo Exercise 5.3 (one rate or two?) with DIC and WAIC instead of a Bayes factor.

Background you need

DIC and WAIC (primer, points 2–3). With a $\text{Uniform}(0, c)$ prior the posteriors are Gamma distributions truncated at c: $\lambda_j \mid \mathbf{Y} \sim \text{Gamma}(Y_j + 1, N_j)$ on $(0, c)$ under M_1, $\lambda_0 \mid \mathbf{Y} \sim \text{Gamma}(Y_1 + Y_2 + 1, N_1 + N_2)$ on $(0, c)$ under M_2.

Key idea

DIC and WAIC depend only on the *posterior*, which barely changes with c (the likelihood is concentrated far below c). The Bayes factor depends on the *prior density* $1/c$ directly.

Solution

Computation.

Draw 50,000 values from each truncated Gamma posterior (inverse-CDF method), evaluate the two Poisson log-likelihood terms for each draw, then apply the DIC and WAIC formulas (each of the two counts is one "observation").

c	Model	$\bar{D}$	p_D	DIC	lppd	p_W	WAIC
1	M_1 separate	14.33	1.91	**16.24**	−6.86	1.00	**15.72**
1	M_2 common	16.44	1.01	17.45	−8.04	0.58	17.24
10	M_1 separate	14.34	1.91	**16.25**	−6.86	1.02	**15.75**
10	M_2 common	16.44	1.00	17.44	−8.04	0.58	17.23

Why? $p_D \approx 2$ and 1: the effective numbers of parameters match the actual numbers, as expected with weak priors.

Answer

Both criteria slightly favour separate rates (M_1): DIC 16.2 vs. 17.4, WAIC 15.7 vs. 17.2 — differences of about 1.2–1.5, too small to be meaningful. Unlike the Bayes factor in Exercise 5.3 (1.40 vs. 0.14), the results are **essentially identical for $c = 1$ and $c = 10$.**

Takeaway & common mistakes

DIC/WAIC behave like cross-validation (predictive accuracy given the posterior) and are robust to vague priors; Bayes factors measure prior predictive accuracy and are not.

Exercise 6.4 — Variable selection with strongly correlated markers

Exercise 4 (exact text from the book)

Consider the regression of the (continuous) CES-D survey score of depression onto p genetic markers. The p covariates for subject i, $X_{i1}, \ldots, X_{ip}$, are aligned along a chromosome giving strong collinearity with the correlation between markers j and k being $\rho^{|j-k|}$ for $\rho = 0.99$.

(a) Say that $p = 3$ and that these generic markers are in fact very important predictors of depression. What do you anticipate the results of Bayesian linear regression with uninformative priors will be?

(b) Write a stochastic search variable selection model for these data, and describe how you would summarize the results.

(c) Assuming the scenario in (a), what do you anticipate the posterior probability of the 2^p models and the p marginal inclusion probabilities will be?

(d) Repeat (d) except with $p = 1{,}000$. How would you summarize these results?

In plain words

Three (or 1,000) almost identical predictors that genuinely matter: predict what regression and variable selection will report, and how to summarise it.

Background you need

- With highly correlated covariates, $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$ is huge for individual coefficients but small for their *sum* (the direction the data identify).
- SSVS: primer of Chapter 5, point 6.

Key idea

The data can tell that *this region* matters, but not *which* of the nearly identical markers carries the signal. To make the predictions concrete we simulated $n = 1{,}000$ subjects with $\rho = 0.99$ and all three true effects equal to 0.3 (noise SD 1).

Solution

(a) Uninformative priors.

Expect wide, strongly negatively correlated posteriors for $\beta_1, \beta_2, \beta_3$, with individual intervals that may include 0 or even have the wrong sign, while the *sum* $\beta_1 + \beta_2 + \beta_3$ is precisely estimated. The simulation confirms this:

	estimate	SE	truth
β_1	0.39	0.22	0.3
β_2	0.60	0.31	0.3
β_3	-0.05	0.23	0.3
$\beta_1 + \beta_2 + \beta_3$	**0.94**	**0.03**	0.9

Not one marker is "significant" on its own, yet together they are overwhelmingly important.

(b) SSVS model.

$Y_i = \beta_0 + \sum_{j=1}^{p} \gamma_j \beta_j X_{ij} + \varepsilon_i$, $\gamma_j \overset{\text{iid}}{\sim} \text{Bernoulli}(q)$, $\beta_j \mid \sigma^2 \sim \text{N}(0, W\sigma^2)$, $\pi(\beta_0) \propto 1$, $\pi(\sigma^2) \propto 1/\sigma^2$ (with $q \sim \text{Beta}$ when p is large). Summaries: posterior model probabilities, marginal inclusion probabilities

$\text{Prob}(\gamma_j = 1 \mid \mathbf{Y})$, and — crucially with collinearity — the probability that *at least one* marker in a correlated block is included.

› (c) Model probabilities for $p = 3$

(exact, $W = 1$):

Model	Probability	Model	Probability
none	0.000	X_3	0.000
X_1	0.066	$X_1 + X_3$	0.048
X_2	0.324	$X_2 + X_3$	0.071
$X_1 + X_2$	**0.403**	$X_1 + X_2 + X_3$	0.089

Inclusion probabilities: X_1 0.61, X_2 0.89, X_3 0.21. The null model has probability 0, so "some marker matters" has probability 1 — but the mass is spread over overlapping subsets, and a truly important marker (X_3) gets a low inclusion probability.

› (d) $p = 1{,}000$.

Expect the same, magnified: the posterior wanders among thousands of near-equivalent models; no single model has appreciable probability; inclusion probabilities are diluted across each correlated block (each marker in a block might get 0.05–0.3 even if the block certainly matters), and results can be sensitive to the MCMC run. Summaries should therefore be at the **region** level:

- sum the inclusion probabilities over windows of adjacent markers (expected number of active markers per region) and plot along the chromosome;
- report *credible sets*: the smallest set of markers that contains an active one with 95% probability (as in genetic fine-mapping);
- report the posterior distribution of the total genetic contribution (e.g. variance explained), which is well identified.

Answer

(a) Individual coefficients wide and negatively correlated (possibly wrong signs), their sum precise. (b) Spike-and-slab SSVS with block-level summaries. (c) Null model probability ≈ 0, but probability spread over several overlapping subsets; marginal inclusion probabilities moderate and uneven (0.61, 0.89, 0.21 in our simulation). (d) Even more dilution; summarise by regions/credible sets rather than individual markers.

Takeaway & common mistakes

Low inclusion probabilities do *not* mean "no effect" when predictors are collinear — always look at blocks.

Exercise 6.5 — Logit or probit for the Gambia data?

Exercise 5 (exact text from the book)

Use DIC and WAIC to compare logistic and probit links for the gambia data in the R package geoR using the five covariates in Listing 6.2 and no random effects.

In plain words

Fit the malaria regression with a logistic and a probit link and compare the fits with DIC and WAIC.

Background you need

Logit link: $\text{Prob}(Y=1) = \frac{e^{\eta}}{1+e^{\eta}}$; probit link: $\text{Prob}(Y=1) = \Phi(\eta)$. Both are S-shaped; the logistic curve has slightly heavier tails. Coefficients are related roughly by $\beta_{\text{logit}} \approx 1.6\,\beta_{\text{probit}}$.

Key idea

Same covariates (age, bed-net use, treated net, greenness, health centre), same priors ($\beta_j \sim \text{N}(0, 100)$ as in the book), only the link differs.

Solution

Fit.

Block Metropolis for each link (20,000 kept draws, acceptance 27–28%); pointwise log-likelihoods $Y_i \log p_i + (1-Y_i)\log(1-p_i)$ for every draw.

Link	$\bar{D}$	p_D	DIC	lppd	p_W	WAIC
logit	2519.4	5.8	2525.2	−1256.8	5.9	2525.3
probit	2521.0	6.1	2527.1	−1257.4	6.2	2527.2

Why? $p_D \approx p_W \approx 6$ = the number of coefficients: the priors are weak.

	logit	probit	ratio
age	0.274	0.167	1.64
netuse	−0.564	−0.349	1.62
treated	−0.289	−0.159	1.81
green	0.295	0.172	1.72
phc	−0.210	−0.134	1.57

Answer

DIC 2525.2 (logit) vs. 2527.1 (probit); WAIC 2525.3 vs. 2527.2. The difference (≈ 2) is negligible: the two links fit equally well, and their coefficients differ only by the usual factor ≈ 1.6. Choose logit for its odds-ratio interpretation.

Takeaway & common mistakes

With probabilities mostly between 0.1 and 0.7, logit and probit curves are almost identical; link choice matters only for extreme probabilities.

Exercise 6.6 — Posterior predictive checks for the Gambia logistic model

Exercise 6 (exact text from the book)

Fit logistic regression model to the gambia data in the previous question and use posterior predictive checks to verify the model fits well. If you find model misspecification, suggest (but do not fit) alternatives.

In plain words

Check whether the logistic model without village effects reproduces key features of the malaria data.

Background you need

PPC: primer, point 4. Children live in 65 villages; a model without village effects assumes that, given the covariates, villages differ only by chance.

Key idea

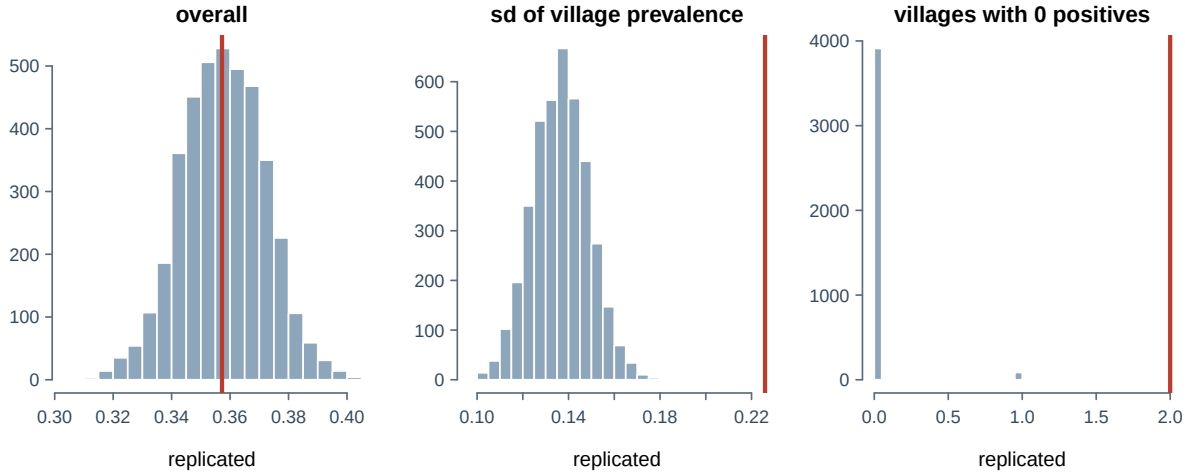
Overall prevalence is fitted by construction. The informative statistics are about *between-village* variation, which the model was not told about.

Solution

› Replicates.

4,000 posterior draws; for each, simulate $Y_i^{\text{rep}} \sim \text{Bernoulli}(p_i^{(s)})$ for all 2,035 children.

Statistic	observed	replicated mean	95% interval	PPP
overall prevalence	0.357	0.358	(0.329, 0.385)	0.52
SD of village prevalence	0.226	0.137	(0.113, 0.161)	**0.000**
max village prevalence	0.933	0.723	(0.600, 0.875)	**0.007**
villages with 0 positives	2	0.02	(0, 0)	**0.000**



By age group, the older groups are reproduced, but the youngest children (under 1.5 years) have lower prevalence than the model predicts (0.19 vs. a replicated 95% interval of 0.21–0.32): the linear age effect is too simple.

› Diagnosis.

The real villages are far more different from each other than the model allows: the SD of village prevalence is 0.23 vs. 0.14 replicated, one village has 93% positives, and two villages have none — something the model essentially never produces.

› Suggested alternatives.

- Village random effects (Exercise 4.8, which estimated $\tau \approx 0.8$).
- Spatially correlated village effects (nearby villages share mosquito habitat), e.g. a Gaussian process in the coordinates.
- A nonlinear age effect (e.g. a spline in age), since the youngest children are over-predicted.

Answer

The model matches overall prevalence but over-predicts malaria in the youngest children and badly under-predicts between-village variation (PPP ≈ 0 for the SD of village prevalence and the number of malaria-free villages). Add village (ideally spatially correlated) random effects and a nonlinear age effect.

Takeaway & common mistakes

Grouping structure that is ignored by the model shows up as extra variation between groups — an easy, targeted PPC.

Exercise 6.7 — Leave-one-out comparison of two clutch models

Exercise 7 (exact text from the book)

For the NBA free throw data in Section 1.6, assume that for player i, $Y_i|p_i \sim \text{Binomial}(n_i, p_i)$ where Y_i is the number of clutch makes, n_i is the number of clutch attempts, and p_i is the clutch make probability. Use leave-one-out cross validation to compare the models:

$$M_1 : \text{logit}(p_i) = \beta_1 + \text{logit}(q_i)$$
$$M_2 : \text{logit}(p_i) = \beta_1 + \beta_2 \,\text{logit}(q_i),$$

where q_i is the overall free throw percentage.

In plain words

Is clutch shooting a constant shift of each player's overall percentage on the logit scale (M_1), or does the shift depend on how good the player is (M_2)? Compare by predicting each player from the other nine.

Background you need

LOO log score: $\sum_{i=1}^{10} \log p(Y_i \mid \mathbf{Y}_{-i})$, with $p(Y_i \mid \mathbf{Y}_{-i}) \approx \frac{1}{S}\sum_s \text{Binomial}(Y_i; n_i, p_i^{(s)})$ using draws from the posterior fitted *without* player i.

Key idea

M_1 is M_2 with β_2 fixed at 1 (a "logit offset"). With only 10 players the extra parameter has to earn its keep.

Solution

Fits.

Priors $\beta_j \sim \text{N}(0, 10^2)$; block Metropolis with GLM-based proposals; 10,000 kept draws per fit. For LOO, each model is refitted 10 times (once without each player).

Player	M_1	M_2	Player	M_1	M_2
Westbrook	−2.45	−2.49	Curry	−1.55	−1.61
Harden	−3.95	−4.00	Antetokounmpo	−2.39	−2.49
Leonard	−2.00	−2.07	Wall	−2.63	−2.62
James	−2.37	−2.84	Davis	−2.32	−2.34
Thomas	−2.02	−2.26	Durant	−1.51	−1.52
Total LOO log score: M_1 **−23.18**, M_2 −24.23 (difference −1.05, SE 0.46)					

Full-data posteriors.

M_1: $\beta_1 = -0.18$ $(-0.39,\ 0.04)$. M_2: $\beta_1 = -0.13$ $(-0.92,\ 0.66)$, $\beta_2 = 0.97$ $(0.49,\ 1.47)$.

Why? M_2 loses mostly on LeBron James and Isaiah Thomas, the players with the lowest and highest overall rates: when one of them is left out, the slope β_2 is poorly determined and predictions for extreme players become unstable.

Answer

LOO-CV prefers the simpler M_1 (log score −23.2 vs. −24.2). Consistently, M_2's slope is $\beta_2 \approx 0.97$, very close to the value 1 that M_1 assumes. Clutch shooting is well described as a common shift of about −0.18 on the logit scale (about 2 percentage points lower in the clutch).

⚠ Takeaway & common mistakes

When the extra parameter's posterior sits at the value the simpler model fixes, CV will (rightly) prefer the simpler model.

Exercise 6.8 — Posterior predictive checks for the clutch model

Exercise 8 (exact text from the book)

Fit model M_2 to the NBA data in the previous question, and use posterior predictive checks to verify the model fits well. If you find model misspecification, suggest (but do not fit) alternatives.

In plain words

Can model M_2 reproduce each player's clutch makes and the overall spread around the fitted rates?

Background you need

Chi-square discrepancy $T(\mathbf{Y}, \theta) = \sum_i \frac{(Y_i - n_i p_i)^2}{n_i p_i (1-p_i)}$: compares observed and replicated spread around the fitted probabilities (detects over-dispersion). Its PPP is $\text{Prob}\{T(\mathbf{Y}^{\text{rep}}, \theta) \geq T(\mathbf{Y}, \theta)\}$.

Key idea

Check both a global statistic (over-dispersion) and each player individually (is any player's count unusual under the model?).

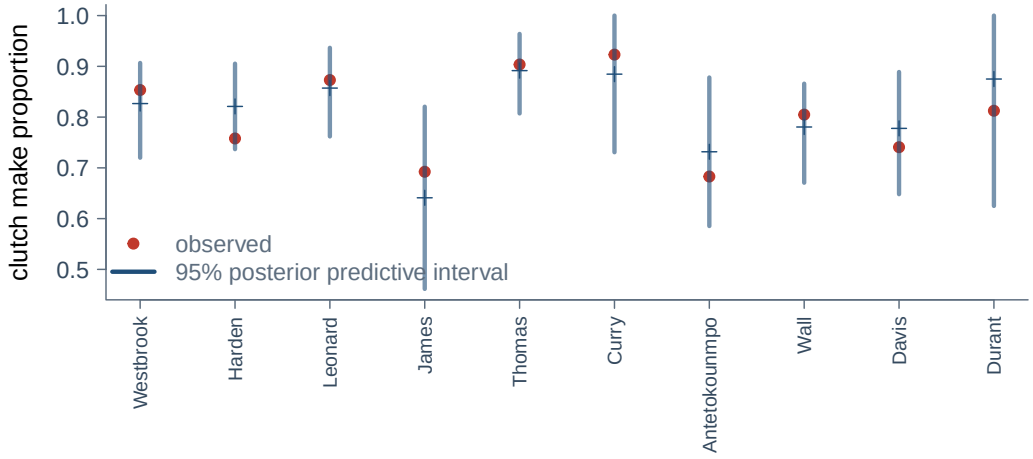
Solution

Global check.

Mean observed discrepancy 8.1 (about what one expects for 10 binomial counts); PPP $= 0.63$: no sign of over-dispersion.

Player-level checks.

Player	observed makes	95% predictive interval	$\text{Prob}(Y^{\text{rep}} \leq Y)$
Westbrook	64	(54, 68)	0.80
Harden	72	(70, 86)	**0.09**
Leonard	55	(48, 59)	0.67
James	27	(18, 32)	0.76
Thomas	75	(67, 80)	0.67
Curry	24	(19, 26)	0.83
Antetokounmpo	28	(24, 36)	0.28
Wall	66	(55, 71)	0.78
Davis	40	(35, 48)	0.35
Durant	13	(10, 16)	0.43



Every observed count lies inside its 95% predictive interval. Harden is at the low edge ($\text{Prob}(Y^{\text{rep}} \leq 72) = 0.09$), consistent with Chapter 1's finding, but with 10 players one value near 0.1 is expected by chance.

› If misfit had appeared.

Add a player-level random effect, $\text{logit}(p_i) = \beta_1 + \beta_2 \text{logit}(q_i) + u_i$ with $u_i \sim \text{N}(0, \tau^2)$ (equivalently a beta-binomial likelihood), to allow genuine individual clutch effects.

Answer

M_2 fits well: chi-square PPP $= 0.63$ and every player's makes fall within their 95% predictive interval. Harden is the most unusual (PPP 0.09) but not alarming. If needed, a player random effect (beta-binomial) would capture individual clutch effects.

Takeaway & common mistakes

Combine a global discrepancy with unit-level checks: global statistics can hide one badly fitted unit.

Exercise 6.9 — Choosing the lag of an autoregression for web traffic

Exercise 9 (exact text from the book)

Download the WWWusage dataset in R. Using data from times $t = 5, \ldots, 100$ as outcomes (earlier times may be used as covariates), fit the autoregressive model

$$Y_t | Y_{t-1}, \ldots, Y_1 \sim \text{Normal}(\beta_0 + \beta_1 Y_{t-1} + \cdots + \beta_L Y_{t-L},\ \sigma^2)$$

where Y_t is the WWW usage at time t. Compare the models with $L = 1, 2, 3, 4$, and select the best time lag L.

In plain words

The number of users connected to a server each minute: predict it from the previous 1–4 minutes and pick how many lags to use.

Background you need

Conditional on the first values, an AR(L) model is a linear regression of Y_t on $(Y_{t-1}, \ldots, Y_{t-L})$, so the Gibbs sampler of Chapter 4 applies. Using the same outcomes ($t = 5, \ldots, 100$) for every L makes the criteria comparable.

Key idea

Compare DIC and WAIC across L (same 96 outcomes, vague priors $\text{N}(0, 100^2)$, $\sigma^2 \sim \text{InvGamma}(0.1, 0.1)$).

Solution

L	$\bar{D}$	p_D	DIC	lppd	p_W	WAIC
1	609.8	3.0	612.8	−303.6	2.6	612.4
2	508.1	4.0	512.1	−252.3	3.7	512.0
3	502.0	5.0	507.0	−248.7	4.9	507.1
4	487.0	5.9	**492.9**	−240.9	5.6	**493.0**

Posterior means for $L = 4$: $\beta_1 = 2.12$, $\beta_2 = -1.81$, $\beta_3 = 1.06$, $\beta_4 = -0.40$ (all 95% intervals exclude 0), $\sigma = 3.06$.

Why? Each extra lag lowers DIC/WAIC by more than 5, and the fourth lag still by 14: the series has smooth, wave-like dynamics that need several lags to describe.

Answer

DIC and WAIC agree: $L = 4$ is best (DIC 492.9, WAIC 493.0), clearly better than $L = 3$ (≈ 507). Select $L = 4$ (within the lags considered).

Takeaway & common mistakes

$\beta_1 \approx 1$ for $L = 1$ and $\sum_l \beta_l \approx 0.98$ for $L = 4$: the series is close to non-stationary, suggesting a model for the *differences* $Y_t - Y_{t-1}$ (Exercise 10).

Exercise 6.10 — Checking the AR(2) model for web traffic

Exercise 10 (exact text from the book)

Using the WWWusage dataset in the previous problem, fit the model with $L = 2$ and use posterior predictive checks to verify that the model fits well. If you find model misspecification, suggest (but do not fit) alternatives.

In plain words

Do the AR(2) model's residuals behave like independent noise?

Background you need

If the model is right, the errors $\varepsilon_t = Y_t - \mu_t$ are independent $\mathrm{N}(0, \sigma^2)$, so their autocorrelation should be near 0. Replicated errors $\varepsilon_t^{\text{rep}} = Y_t^{\text{rep}} - \mu_t$ show what "near 0" means for 96 points.

Key idea

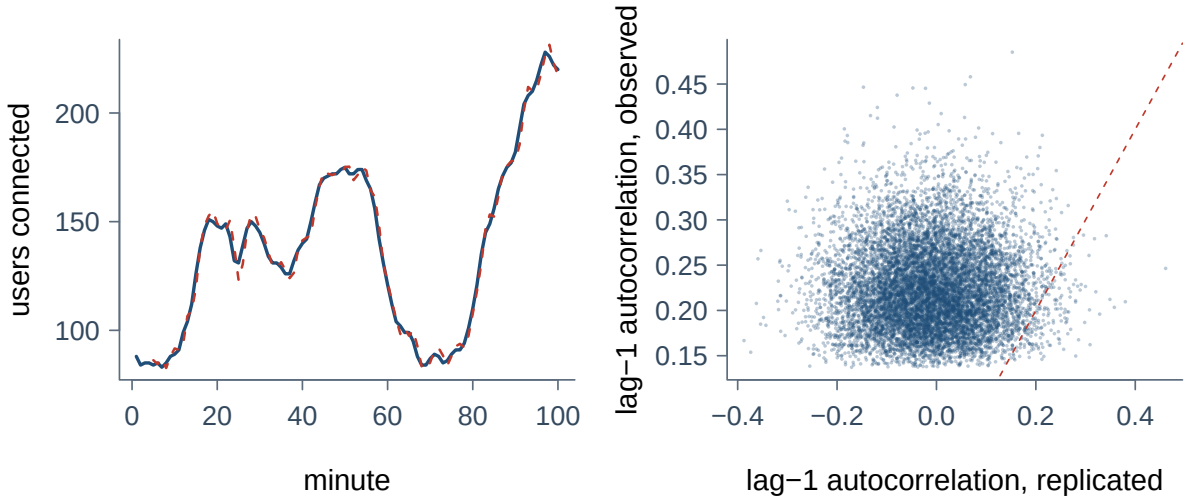
For time series, the best PPC statistics are residual autocorrelations at lags 1 and 2, plus kurtosis and the largest absolute residual.

Solution

Statistics

(averaged over posterior draws; PPP $= \mathrm{Prob}(T^{\text{rep}} \geq T^{\text{obs}})$):

	lag-1 ACF	lag-2 ACF	kurtosis	max $\lvert\varepsilon\rvert$
observed	0.225	−0.314	−0.32	8.5
replicated	−0.010	−0.009	−0.12	9.3
PPP	**0.016**	**0.993**	0.63	0.67



The residuals are clearly autocorrelated (positive at lag 1, negative at lag 2), which independent noise essentially never produces; tails/extremes are fine.

Suggested alternatives.

- More lags ($L = 3, 4$ — as Exercise 9 already found).

- Model the differences: an AR model for $\Delta Y_t = Y_t - Y_{t-1}$ (an ARIMA$(p, 1, 0)$), since the series behaves like an integrated process (the differences have lag-1 autocorrelation 0.79).
- Possibly time-varying variance (the series has calm and volatile stretches).

Answer

The AR(2) model fails the residual-autocorrelation checks (lag-1 PPP 0.016, lag-2 PPP 0.993), so it is misspecified; kurtosis and extremes are fine. Use more lags or, better, an AR model for the first differences (ARIMA).

Takeaway & common mistakes

For time series, "independent errors" is the assumption most worth checking.

Exercise 6.11 — A prior that makes p_D uniform

Exercise 11 (exact text from the book)

For the one-way random effects model in (6.12), explain how you would select a prior for c so the p_D has roughly a uniform prior on p_D.

In plain words

In the random-effects model, the effective number of parameters p_D depends on the prior variance ratio c. Choose a prior for c that makes every model size between 0 and p equally likely.

Background you need

Model (6.12): $Y_{ij} \mid \mu_j \sim \mathrm{N}(\mu_j, \sigma^2)$, $\mu_j \sim \mathrm{N}(0, c\sigma^2)$, n replicates in each of p groups. The book shows $p_D = p\,\frac{c}{c+1/n}$. **Change of variables**: if $U = g(c)$ and we want $U \sim \mathrm{Uniform}(0,1)$, then $c = g^{-1}(U)$ and $\pi(c) = |g'(c)|$.

Key idea

Make $U = p_D/p = \frac{c}{c+1/n} = \frac{nc}{1+nc}$ uniform on $(0,1)$ and solve for c.

Solution

› Step 1 — invert.

$U = \frac{nc}{1+nc} \iff U + Unc = nc \iff c = \frac{U}{n(1-U)}$.

› Step 2 — the implied density.

$\frac{\mathrm{d}U}{\mathrm{d}c} = \frac{n(1+nc) - nc\cdot n}{(1+nc)^2} = \frac{n}{(1+nc)^2}$, so

$$\pi(c) = \frac{n}{(1+nc)^2}, \qquad c > 0.$$

This is proper ($\int_0^\infty \frac{n}{(1+nc)^2}\,\mathrm{d}c = 1$), with median $1/n$ and a heavy right tail. Equivalently nc has the "beta-prime$(1,1)$" (log-logistic) distribution.

› Step 3 — in practice.

Either use this density directly, or put a $\mathrm{Uniform}(0,1)$ prior on U (the "shrinkage fraction") and define $c = U/\{n(1-U)\}$ in the model code. If group sizes differ, the same idea applies with n replaced by a typical group size (making p_D approximately uniform).

Sanity check

Simulation with $n = 5$, $p = 10$: 10^5 draws of c give p_D with mean 4.997 (theory 5) and a KS distance of 0.002 from $\mathrm{Uniform}(0, 10)$; the median of c is $0.199 \approx 1/5$. ✓

Answer

Set $c = \dfrac{U}{n(1-U)}$ with $U \sim \mathrm{Uniform}(0,1)$, i.e. $\pi(c) = \dfrac{n}{(1+nc)^2}$; then $p_D/p = \frac{c}{c+1/n} = U$ is uniform, so $p_D \sim \mathrm{Uniform}(0, p)$.

⚠ Takeaway & common mistakes

Putting priors on interpretable quantities ("how much shrinkage", "effective model size") is a principled way to choose priors for variance parameters.

Exercise 6.12 — How many spline basis functions for the temperature curve?

Exercise 12 (exact text from the book)

The global mean temperature anomaly (difference between current year and the 1951-1980 average) data on the course website are downloaded from https://data.giss.nasa.gov/gistemp/ and plotted below. Let X_i and Y_i be the year and temperature anomaly (C) for observation i. We will analyze these data using a nonparametric regression model $Y_i = g(X_i) + \varepsilon_i$ as in Section 8.3, where g is modeled as a linear combination of J B-spline basis functions.

(a) Select J using DIC and WAIC.

(b) Select J using cross-validation.

(c) For the selected value of J, check model fit using posterior predictive checks.

In plain words

Choose how flexible the temperature curve of Exercise 4.10 should be, three ways, and then check the chosen fit.

Background you need

B-spline regression (Exercise 4.10): $g(x) = \sum_{j=1}^{J} \beta_j B_j(x)$, a linear regression on J basis columns. More basis functions = more wiggly curve.

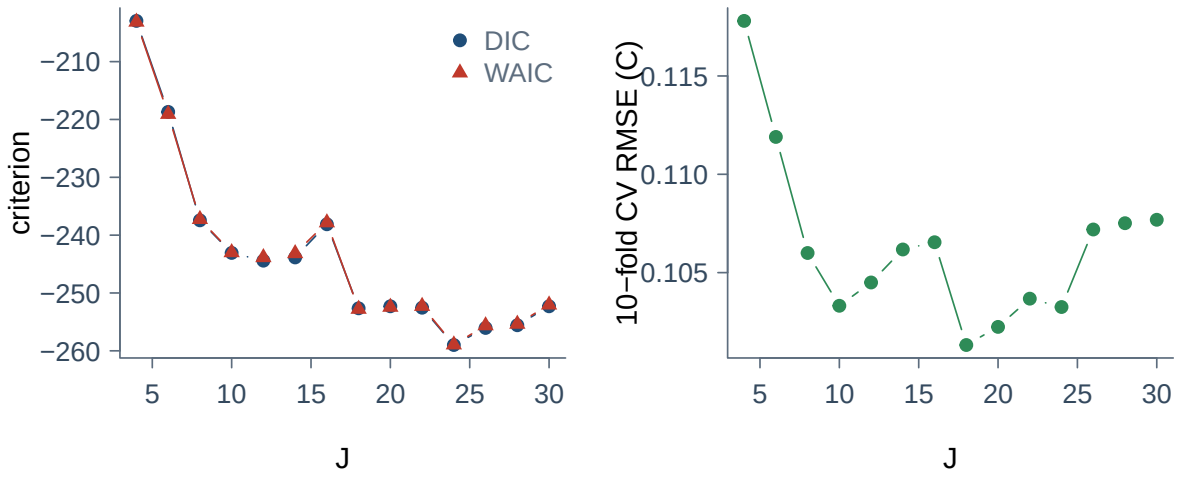
Key idea

Fit $J = 4, 6, \ldots, 30$ (priors $\beta_j \sim \mathrm{N}(0, 100^2)$, $\sigma^2 \sim \mathrm{InvGamma}(0.01, 0.001)$ — vague on the scale of these residuals, whose variance is about 0.01); compute DIC and WAIC from the posterior, and 10-fold CV prediction error. Then run PPCs on residual autocorrelation (yearly temperatures are persistent).

Solution

(a), (b) Criteria.

J	DIC	p_D	WAIC	p_W	10-fold CV RMSE (°C)
4	−203.0	4.9	−203.1	4.5	0.118
8	−237.4	9.0	−237.3	8.5	0.106
10	−243.1	11.0	−243.0	10.1	0.103
12	−244.4	12.9	−243.9	12.2	0.105
16	−238.1	17.0	−237.8	15.4	0.107
18	−252.7	18.6	−252.8	16.5	**0.101**
20	−252.3	21.1	−252.4	18.5	0.102
24	**−259.0**	25.1	**−258.9**	21.9	0.103
26	−256.0	26.9	−255.6	23.6	0.107
30	−252.3	30.8	−252.0	26.5	0.108



DIC and WAIC are minimised at $J = 24$; 10-fold CV at $J = 18$. All three criteria are nearly flat for J between about 10 and 24 (CV RMSE varies by only 0.002°C), so any J in that range is defensible; the zig-zag is partly Monte Carlo and fold-to-fold noise.

› (c) PPCs for $J = 24$.

	lag-1 ACF of residuals	kurtosis	max $\lvert\varepsilon\rvert$	sign changes
observed	0.235	−0.53	0.223	61.6
replicated	−0.007	−0.08	0.255	70.5
PPP	**0.005**	0.87	0.77	0.91

The residuals are positively autocorrelated (warm years follow warm years beyond the smooth trend; e.g. El Niño episodes) and change sign less often than independent noise would. Marginal shape (kurtosis, extremes) is fine.

Answer

(a) DIC and WAIC choose $J = 24$. (b) 10-fold CV chooses $J = 18$; the criteria are flat for $J \approx 10$–24. (c) The selected fit fails the residual-autocorrelation check (PPP 0.005): add AR(1) errors (or a smoother with a penalty) for honest uncertainty; the curve itself is fine.

Takeaway & common mistakes

When criteria are flat over a range, report that range — and remember that autocorrelated errors make DIC/WAIC/CV favour overly wiggly curves.

Exercise 6.13 — t vs. Gaussian for the galaxies: posterior predictive checks

Exercise 13 (exact text from the book)

Open and plot the galaxies data in R using the code below,

```
> library(MASS)
> data(galaxies)
> ?galaxies
> Y <- galaxies
> n <- length(Y)
> hist(Y,breaks=25)
```

Model the observations $Y_1, \ldots, Y_{82}$ using the Student's t-distribution with location μ, scale σ and degrees of freedom k. Assume prior distributions $\mu \sim \text{Normal}(0, 10000^2)$, $1/\sigma^2 = \tau \sim \text{Gamma}(0.01, 0.01)$ and $k \sim \text{Uniform}(1, 30)$.

(a) Use posterior predictive checks to evaluate whether the t distribution captures the mean, variance, skewness and kurtosis of the data.

(b) Repeat this analysis with a Gaussian model.

In plain words

Fit a t and a normal distribution to the galaxy velocities and check whether replicated data match the observed mean, variance, skewness and kurtosis.

Background you need

Student-t Gibbs sampler (Exercise 3.9). Sample skewness $\frac{\frac{1}{n}\sum(Y_i-\bar{Y})^3}{s^3}$ and excess kurtosis $\frac{\frac{1}{n}\sum(Y_i-\bar{Y})^4}{s^4} - 3$ (both 0 for a normal distribution).

Key idea

A PPC passes when the observed statistic is a typical value of the replicated statistic (PPP not close to 0 or 1).

Solution

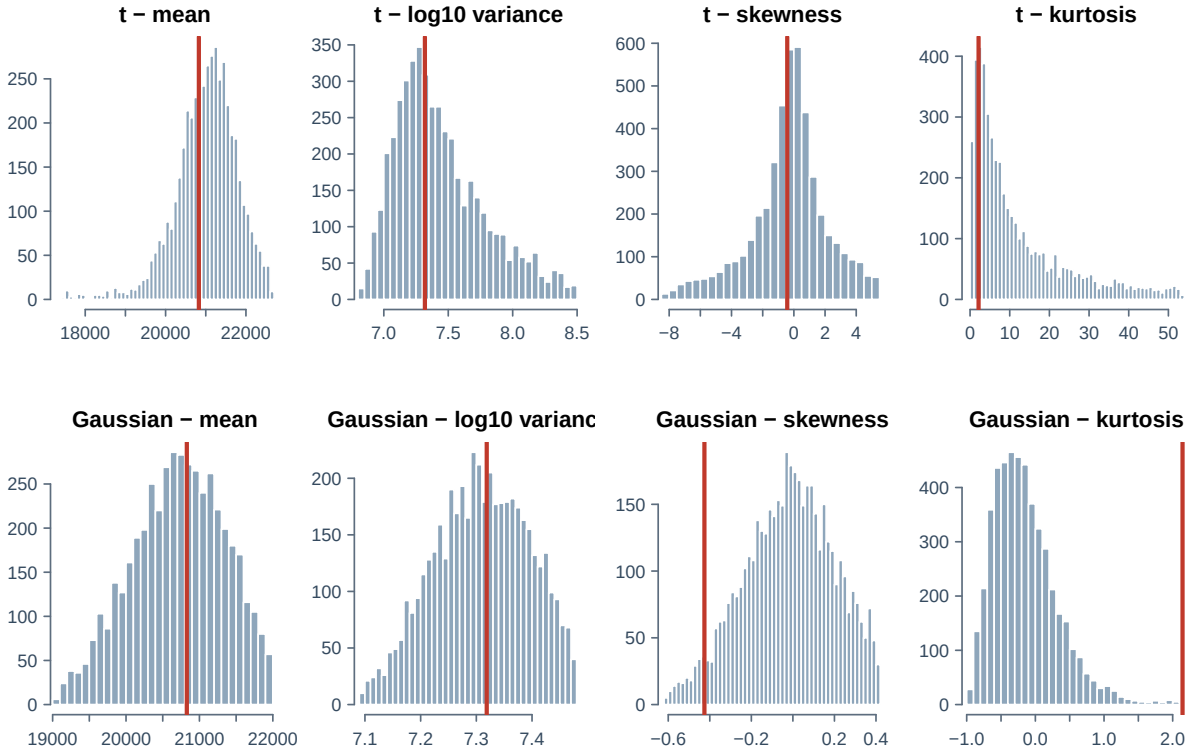
Fits.

t model: posterior as in Exercise 3.9 ($\mu \approx 21{,}100$, $\sigma \approx 2{,}240$, $k \approx 2.1$). Gaussian model (same priors for μ, τ): $\mu = 20{,}776$ (19,764–21,781), $\sigma = 4{,}603$ (3,952–5,395).

Checks

(5,000 replicated datasets of 82 values each):

Statistic	observed	t model 95% replicated interval	t model PPP	Gaussian model 95% replicated interval	Gaussian model PPP
mean	20,828	(18,870, 23,280)	0.65	(19,310, 22,210)	0.47
variance	2.08×10^7	$(0.8 \times 10^7,\ 67 \times 10^7)$	0.58	$(1.3 \times 10^7,\ 3.2 \times 10^7)$	0.49
skewness	-0.43	$(-6.9,\ 6.7)$	0.59	$(-0.51,\ 0.49)$	0.95
kurtosis	2.14	(0.46, 64.9)	0.85	$(-0.85,\ 1.02)$	**0.002**



Answer

(a) The t model passes all four checks (PPP 0.58–0.85) — but only because with $k \approx 2$ its replicated variance, skewness and kurtosis are so variable that almost anything is "typical". (b) The Gaussian model matches the mean and variance but fails kurtosis badly (PPP 0.002: the data have far heavier tails than a normal), and skewness is borderline (PPP 0.95).

Takeaway & common mistakes

Passing PPCs is not proof of a good model: the t "passes" by being vague, yet Exercises 3.9 and 4.13 showed it misses the multi-modality. Add statistics that target the suspected failure (e.g. the number of modes or the gap between clusters).

Exercise 6.14 — Strengths and weaknesses of model-selection tools

Exercise 14 (exact text from the book)

For each model selection approach below, give an advantage and a limitation of the approach and a hypothetical situation where it would be a good option.

(a) Bayes factors

(b) Stochastic search variable selection

(c) Cross validation

(d) DIC/WAIC

In plain words

Compare four model-selection tools: what each does well, what it does badly, and when to use it.

Key idea

Bayes factors and SSVS ask "which model is *true*?" (prior predictive); CV and DIC/WAIC ask "which model *predicts* best?" (posterior predictive).

Solution

Approach	Advantage	Limitation	Good situation
(a) Bayes factors	Coherent: gives posterior model probabilities and can support the null; automatically penalises complexity.	Very sensitive to vague priors (Lindley's paradox, Exercise 5.3); marginal likelihoods can be hard to compute.	Comparing a few scientifically motivated models with informative priors, e.g. "does a drug have an effect?" with a prior based on earlier trials.
(b) SSVS	Explores huge model spaces (2^p) and gives inclusion probabilities and model-averaged predictions.	Inherits BF prior sensitivity; mixes poorly and dilutes probabilities with collinear predictors (Exercise 4).	Choosing among $p = 10$–100 candidate predictors, e.g. which neighbourhood variables affect house prices (Exercise 5.9).
(c) Cross validation	Directly measures out-of-sample prediction; few assumptions; works for any model, even non-Bayesian ones.	Computationally expensive (refit K times); noisy for small n; needs care with dependent data (time series, clusters).	Choosing a predictive model, e.g. which covariates best forecast ozone (Exercise 1).
(d) DIC / WAIC	Cheap (one MCMC run); estimate predictive accuracy and effective model size.	Approximations; DIC can misbehave (negative p_D, multimodal posteriors); both assume roughly independent observations.	Quickly comparing several fitted hierarchical models, e.g. random-effect distributions for the Gambia data (Section 6.4).

Answer

See the table: Bayes factors — coherent but prior-sensitive (use with informative priors); SSVS — searches many models but struggles with collinearity (use for moderate p variable selection); CV — assumption-light prediction measure but costly (use for predictive model choice); DIC/WAIC — cheap predictive approximations (use to compare hierarchical models fitted by MCMC).

⚠ Takeaway & common mistakes

Decide first whether the goal is *explanation* (which model is true?) or *prediction* (which model predicts best?) — that decides the tool.